

Covariance, correlation and the bivariate normal

One number for how two variables move together, what it misses, and the bell-shaped cloud it describes fully.

THE QUESTION

Do two random variables rise and fall together, and how strongly?

THE INTUITION

Multiply each pair's distances from the two means and average. Pairs on the same side of both means push the average up; pairs on opposite sides pull it down. Divide by both SDs to get a unit-free ρ between -1 and 1 .

KEY TAKEAWAY

$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$, $\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y)$.
Independent gives $\rho = 0$; $\rho = 0$ does not give independent. ρ only sees straight-line trends.

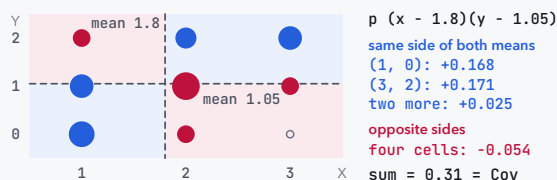
$E[XY] \rightarrow$ minus $E[X]E[Y]$: Cov $\rightarrow$ divide by both SDs: $\rho \rightarrow \rho = 0$? look at the shape $\rightarrow$ bivariate normal $\rightarrow$ predict with ρ

1 Covariance is $E[XY]$ minus $E[X]E[Y]$

	y = 0	y = 1	y = 2	
x = 1	0 · 0.20	1 · 0.15	2 · 0.05	$E[XY] = 2.20$
x = 2	0 · 0.05	2 · 0.25	4 · 0.10	$E[X] = 1.80$
x = 3	0 · 0.00	3 · 0.05	6 · 0.15	$E[Y] = 1.05$
				$E[X]E[Y] = 1.89$
				$\text{Cov} = 0.31 = 2.20 - 1.89$

$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$; expanding gives $E[XY] - E[X]E[Y]$. Each cell of the shop table (the joint distributions sheet) adds $xy \cdot p(x, y)$: $E[XY] = 2.2$ against $1.8 \times 1.05 = 1.89$, so $\text{Cov} = 0.31$ and bigger baskets take more bags. $\text{Cov}(X, X) = \text{Var } X$. Independent variables have $E[XY] = E[X]E[Y]$ and zero covariance: two independent $U(0, 1)$ give $1/4 = 1/2 \cdot 1/2$.

2 The sign says which quadrants hold the mass



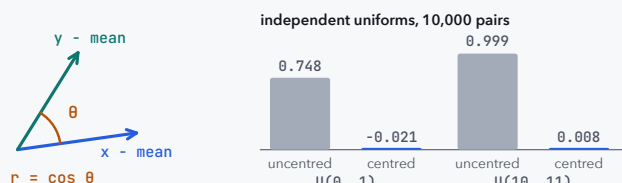
Mass up-right or down-left of the means adds $p(x - \mu_X)(y - \mu_Y) > 0$; mass on the other diagonal subtracts. Here $0.364 - 0.054 = 0.31$, as the shortcut said. Covariance is bilinear: $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$ and $\text{Cov}(A + B, C) = \text{Cov}(A, C) + \text{Cov}(B, C)$.

3 Correlation rescales covariance to $[-1, 1]$



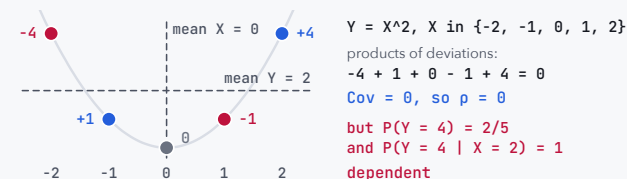
$\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y)$ is the covariance of the standardised variables, so units cancel. Shop: $0.31 / (0.748 \times 0.740) = 0.56$. $|\rho| = 1$ only when $Y = aX + b$ exactly; ρ says how tightly points hug a line, not how steep it is. Each cloud: 200 simulated pairs (seed 0).

4 Correlation is the cosine of an angle



With the n centred values as vectors, $r = (x - \bar{x}) \cdot (y - \bar{y}) / (\|x - \bar{x}\| \|y - \bar{y}\|) = \cos \theta$: 1 for the same direction, 0 at right angles, -1 for opposite ones. Centring matters: $E[XY] / \sqrt{E[X^2]E[Y^2]}$ is $3/4$ for independent $U(0, 1)$ and 0.9992 for $U(10, 11)$ as both sit far from 0. Centred, both are near 0.

5 Uncorrelated does not mean independent



X is $-2, \dots, 2$ with chance $1/5$ each and $Y = X^2$. $E[X] = 0$ and $E[XY] = E[X^3] = 0$, so $\text{Cov} = 0$, yet X fixes Y : the U shape cancels. Independent always gives $\rho = 0$; the reverse holds for the bivariate normal (page 2), not in general.

6 A high correlation does not show a cause

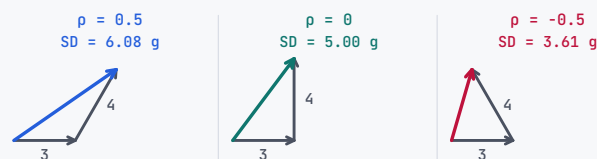


On six made-up days, ice creams sold and sunburn cases have $r = 0.987$. Both follow temperature (25 to 40°C ; $r = 0.998$ and 0.995), a confounder. Selling fewer ice creams would not prevent a sunburn. Compare days of equal temperature, or run an experiment.

Adding variables and the bivariate normal

Covariance enters the variance of every sum, and for the bivariate normal five numbers fix the whole shape.

7 $\text{Var}(X \pm Y) = \text{Var } X + \text{Var } Y \pm 2 \text{Cov}$



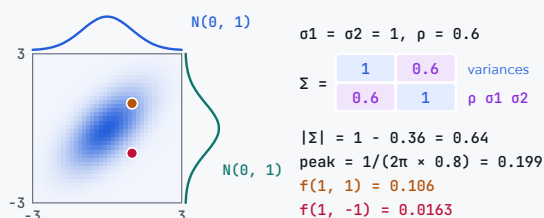
Expand $\text{Cov}(X + Y, X + Y)$ by bilinearity; in general $\text{Var}(aX + bY) = a^2 \text{Var } X + b^2 \text{Var } Y + 2ab \text{Cov}$. Independent 3 g and 4 g scale errors give $\sqrt{9 + 16} = 5$ g, not 7: variances add, SDs do not. SDs add like arrows at angle θ with $\cos \theta = \rho$. Shop: $0.56 + 0.5475 + 2 \times 0.31 = 1.7275$, the variance of $X + Y$ from its own PMF.

8 Estimate them with `np.cov` and `np.corrcoef`

```
rng = np.random.default_rng(0)
S = [[1, 0.5], [0.5, 1]]
xy = rng.multivariate_normal(
    [0, 0], S, 1000)
x, y = xy.T
np.cov(x, y)[0, 1] # 0.539
np.corrcoef(x, y)[0, 1] # 0.528
np.cov(100*x + 7, y)[0, 1] # 53.9
z = rng.standard_normal(1000)
np.corrcoef(z, z**2)[0, 1] # 0.030
```

Both return 2×2 matrices with the variances on the diagonal; `np.cov` divides by $n - 1$. Rescaling x by 100 scales the covariance by 100 and leaves r . z and z^2 are dependent, yet $r \approx 0$.

9 Five numbers fix a bivariate normal

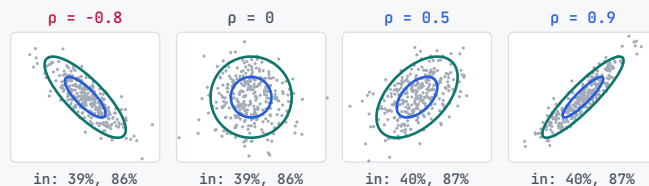


With $z_1 = (x - \mu_1)/\sigma_1$ and $z_2 = (y - \mu_2)/\sigma_2$,

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{z_1^2 - 2\rho z_1 z_2 + z_2^2}{2(1-\rho^2)}\right).$$

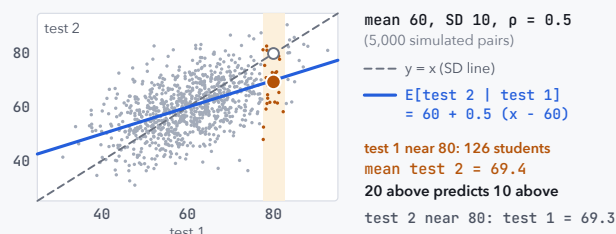
Derivation: two independent normals multiply, with exponent $-\frac{1}{2}d^T \Sigma^{-1}d$, $d = x - \mu$ and diagonal Σ . Allowing $\Sigma_{12} = \rho\sigma_1\sigma_2$ gives $|\Sigma| = \sigma_1^2\sigma_2^2(1-\rho^2)$ and $f = \exp(-\frac{1}{2}d^T \Sigma^{-1}d)/(2\pi\sqrt{|\Sigma|})$ for any symmetric positive-definite Σ . The marginals stay normal.

10 The contours are ellipses tilted by ρ



Where the exponent is constant, so is f : the contours are ellipses around μ . With equal SDs they are circles at $\rho = 0$, tilt up-right for $\rho > 0$ and flatten onto a line as $\rho \rightarrow \pm 1$. The ellipses $d^T \Sigma^{-1}d = 1$ and 4 hold $1 - e^{-1/2} = 39\%$ and $1 - e^{-2} = 86\%$ at every ρ (shares of 10,000 draws). Here $\rho = 0$ makes f factor, so uncorrelated means independent. The axes follow the eigenvectors of Σ (the random vectors and PCA sheet).

11 Regression to the mean comes from $|\rho| < 1$



Given $X = x$, Y is normal with mean $\mu_Y + \rho(\sigma_Y/\sigma_X)(x - \mu_X)$ and variance $\sigma_Y^2(1 - \rho^2)$. In standard units the prediction is ρz : 20 points above the mean on test 1 predicts 10 above on test 2 at $\rho = 0.5$. It works both ways, so nobody got worse; extreme scores held some luck.

12 Pause and think

$\text{Cov} = 2$, $\text{Var } X = 4$, $\text{Var } Y = 9$: $\rho? 2/(2 \times 3) = 1/3$. $\text{Var } X = 1$, $\text{Var } Y = 4$, $\text{Cov} = 1$: $\text{Var}(2X - 3Y)? 4 + 36 - 12 = 28$.

Can $\text{Cov} = 5$ when $\text{Var } X = \text{Var } Y = 4$? No: $|\text{Cov}| \leq \sigma_X \sigma_Y = 4$.

Temperature in $^\circ\text{F}$ instead of $^\circ\text{C}$: does ρ change? No. The covariance grows by 1.8.

$\rho(X, -X)$ and $\rho(X, 2X + 3)? -1$ and 1, whatever the slope.

A group 2 SD below the mean, $\rho = 0.5$: next test? 1 SD below on average; they seem to improve.

$$\text{Cov} = E[XY] - E[X]E[Y] \cdot \rho = \text{Cov}(X, Y)/(\sigma_X \sigma_Y) \text{ in } [-1, 1] \cdot \text{independent gives } \rho = 0, \text{ not back} \cdot \text{Var}(X + Y) = \text{Var } X + \text{Var } Y + 2 \text{Cov} \cdot \text{predict } \rho z$$

Source: the course notebooks "Joint distribution properties" ($E[XY]$ and $\cos \theta$ for independent uniforms) and "Distributions in 2D", the slides "Derivation of bivariate normal distribution" and the normal-shape app, nipunbatra.github.io/psdv-teaching. The shop table, the ice-cream days and the test scores are our own illustrative numbers; all numbers come from the sheet's figure script (seed 0). Next: the sums of random variables, LLN and CLT sheet.